

Predicting elbow varus torque in pitchers a model has never seen

How much of a pitcher's elbow load can be predicted from body size, velocity and mechanics, and how much does leaky validation overstate it?

Andres Perez, M.S. Kinesiology
(Biomechanics)

Python · statsmodels · scikit-learn

Code: github.com/Andresperez397/obp-elbow-torque

DATA

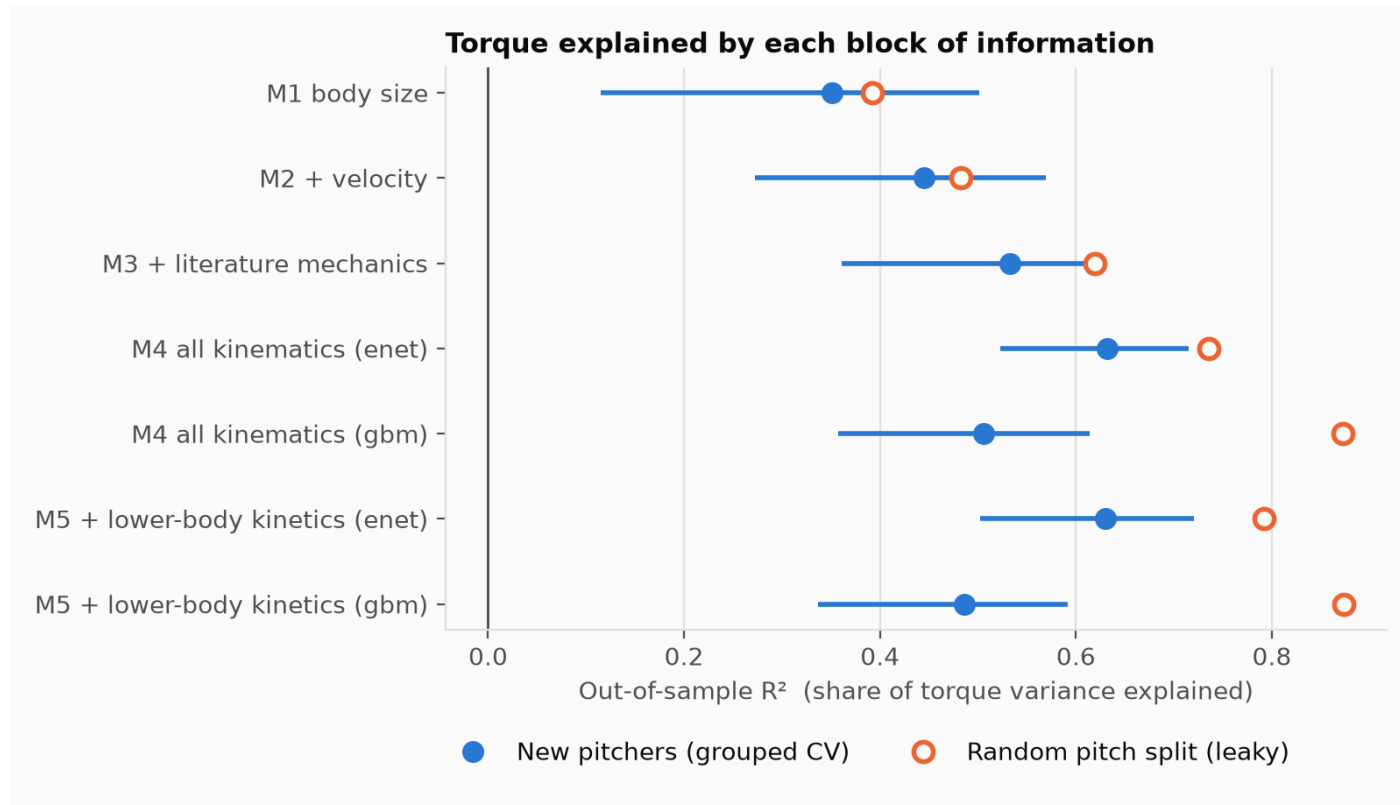
411 fastballs from 100 pitchers in the OpenBiomechanics Project (Driveline Baseball): lab motion capture with inverse dynamics. The pitchers are mostly college level, with one session each. The outcome is peak elbow varus torque (Nm).

APPROACH

- **Pre-registered:** the questions, models and decision rules were committed to git before any model was fit.
- **No leakage:** predictors from the same arm inverse-dynamics solution were excluded.
- **Grouped validation:** by pitcher, 10 folds $\times$ 20 repeats; all preprocessing done inside folds.
- **Uncertainty:** pitcher-level bootstrap CIs; mixed models for the variance split and velocity effects.

FINDINGS

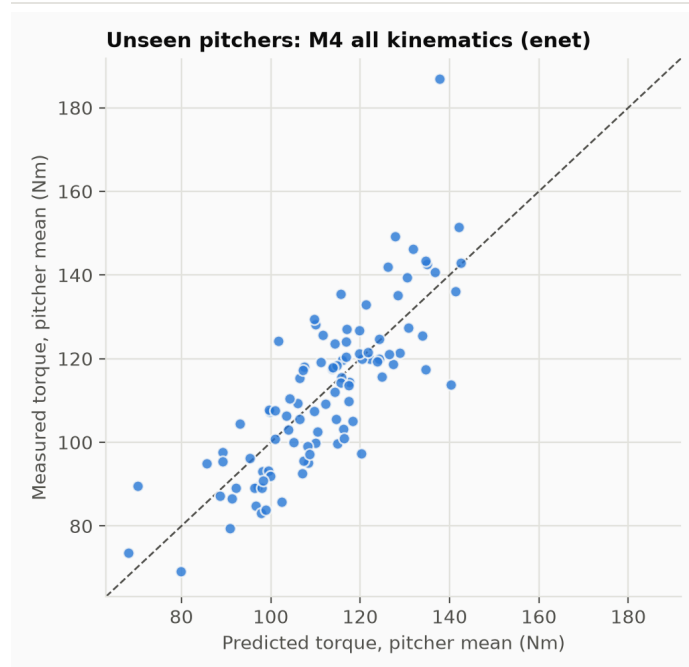
- **Torque is a pitcher trait:** 95% of the variance lies between pitchers (ICC 0.95).
- **Mechanics matter:** on unseen pitchers, all kinematics explain R^2 0.63 (CI 0.52–0.72), against 0.35 for body size alone.
- **Lower-body kinetics and ground reaction forces add nothing** beyond kinematics.
- **Leaky validation misleads:** random pitch splits made boosted trees look best (R^2 0.87). On new pitchers they reach 0.51, below the linear model.
- **Velocity cost:** about 1.3 Nm per mph, both between and within pitchers.



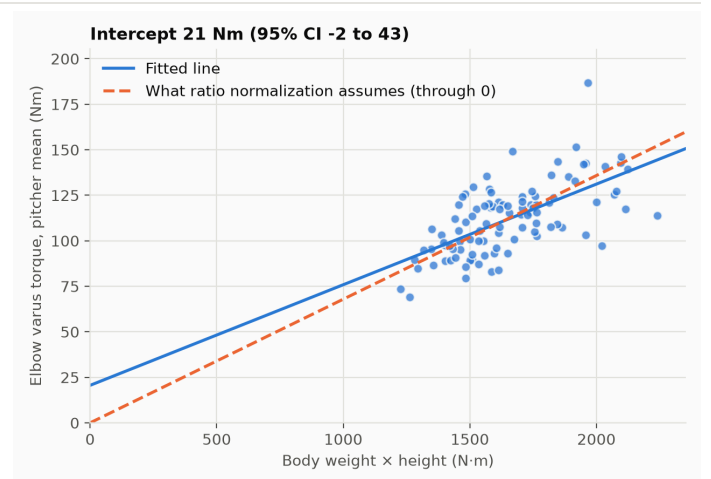
Out-of-sample R^2 for each information block. Filled: validated on held-out pitchers, with 95% CI. Hollow: the same model under a random pitch split.

MODEL RESULTS (HELD-OUT PITCHERS)

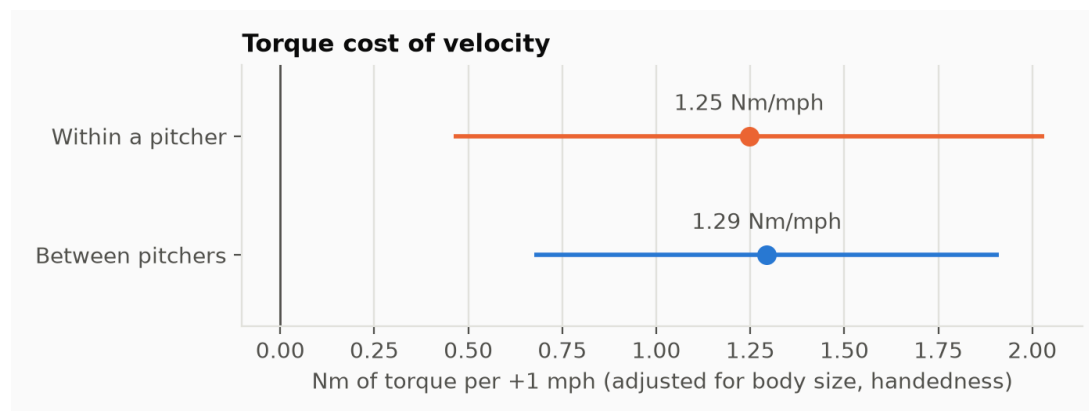
Model	R ²	95% CI	RMSE (Nm)	R ² , leaky split
Body size + handedness	0.35	0.12–0.50	16.0	0.39
+ velocity	0.45	0.27–0.57	14.8	0.48
+ 10 literature mechanics	0.53	0.36–0.63	13.6	0.62
All kinematics, elastic net	0.63	0.52–0.72	12.0	0.74
All kinematics, boosted trees	0.51	0.36–0.61	14.0	0.87
+ lower-body kinetics, elastic net	0.63	0.50–0.72	12.1	0.79



Pitcher-mean torque against the prediction for pitchers held out of training.



Ratio normalization assumes a line through zero. The intercept CI includes zero, but only barely.



LIMITATIONS

The sample is 100 mostly college pitchers from one facility, throwing fastballs in a lab, so results may not transfer to MLB or in-game settings. Torque is an inverse-dynamics estimate whose level scales with body mass by construction. With 100 pitchers, R² gains under about 0.05 can't be reliably separated from zero.

Data: The OpenBiomechanics Project, Driveline Baseball (CC BY-NC-SA 4.0). This is a derived work under the same license and is not affiliated with or endorsed by Driveline.